Lesson 14A Density and Pressure

Introduction to Fluids: Density and Pressure

The study of fluids is foundational to many areas of physics and engineering, including hydraulics, meteorology, and biomechanics. A **fluid** is any substance that can flow and take the shape of its container—this includes both **liquids**(like water) and **gases** (like air).

What makes a fluid distinct is that it cannot resist shear stress (forces parallel to its surface). Instead, fluids can only exert forces **perpendicular** to surfaces—this is what leads to the pressure in a fluid. Even materials like pitch, which flow extremely slowly, are still considered fluids because they lack a crystalline structure and can conform to a container over time.

Density

The **density** of a fluid is a measure of how much mass is packed into a given volume. It's defined as:

$$\rho = \frac{m}{V}$$

In theory, we define density at a point as the limit of mass per volume for an infinitesimally small sample. But in practice, we assume that fluid samples are large compared to atomic scales and have a uniform density:

$$\rho = \text{constant throughout the sample}$$

- Units: $\mathrm{kg/m^3}$
- Density is a **scalar** quantity

Pressure

Pressure in a fluid is defined as the **force per unit area** exerted perpendicular to a surface:

$$p = \frac{F}{A}$$

In the general case, pressure is the limit of force over an infinitesimally small area. But for a flat surface with uniform force, we simply use the above formula. This is measured directly with pressure sensors.

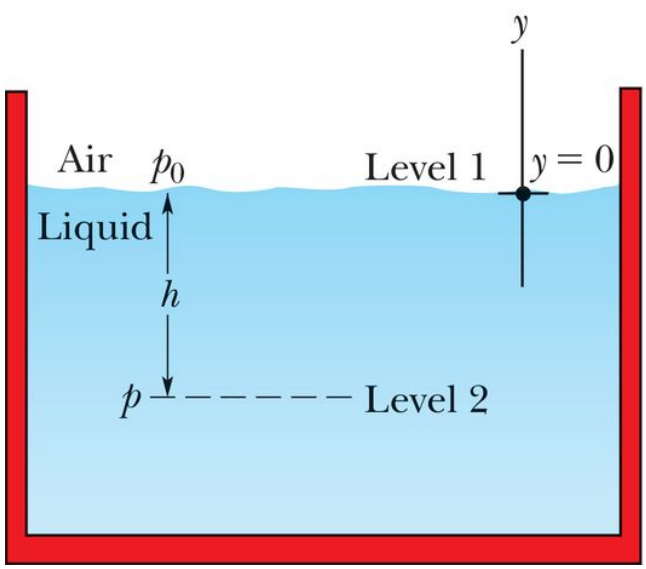
Experiments show that in a **fluid at rest**, the pressure at a given point is **the same in all directions**—it is **scalar**, even though the force it causes is a vector.

Units of Pressure

- **Pascal (Pa)**: $1\,\mathrm{Pa} = 1\,\mathrm{N/m^2}$
- **Atmosphere (atm)**: $1\,\mathrm{atm} = 101{,}325\,\mathrm{Pa}$
- **Torr**: $1\,\mathrm{torr} = 1\,\mathrm{mm\ Hg}$
- **Pounds per square inch (psi)**: Common in engineering applications

Fluids at Rest: Hydrostatics

In a fluid at rest (like water in a container or air in the atmosphere), the pressure **increases with depth**. This is because the fluid above exerts weight on the fluid below.



Start by considering a fluid element at depth:

$$\text{Upward force} = \text{Downward force} + \text{Weight of fluid}$$

Substituting pressure and mass with density and area:

$$p = p_0 + \rho g h$$

- p: **absolute pressure** at depth
- p_0: **pressure at the surface** (often atmospheric pressure)
- ρ: fluid density
- g: gravitational acceleration
- h: depth below the surface

Gauge vs. Absolute Pressure

The pressure measured by most devices is the **gauge pressure**, which is the difference between the absolute pressure and atmospheric pressure:

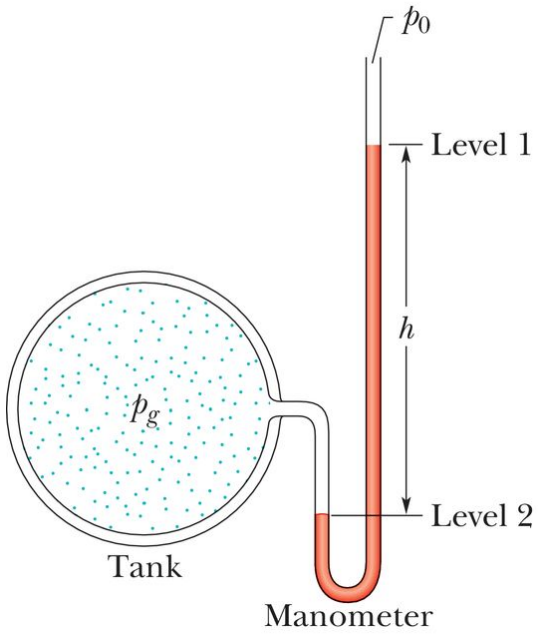
$$p_{\text{gauge}} = p - p_0$$

This means a gauge will read **zero** at atmospheric pressure.

Barometers and Manometers

- A **mercury barometer** measures atmospheric pressure by balancing it against the weight of a column of mercury. The height h of the column corresponds to pressure.
- An **open-tube manometer** compares a gas pressure to atmospheric pressure. The pressure difference is related to the height difference h between the two fluid levels:

$$p_{\text{gauge}} = \rho g h$$

This value can be **positive or negative**, depending on whether the gas pressure is above or

below atmospheric pressure.